

Lanezy

1. Login-Signup:

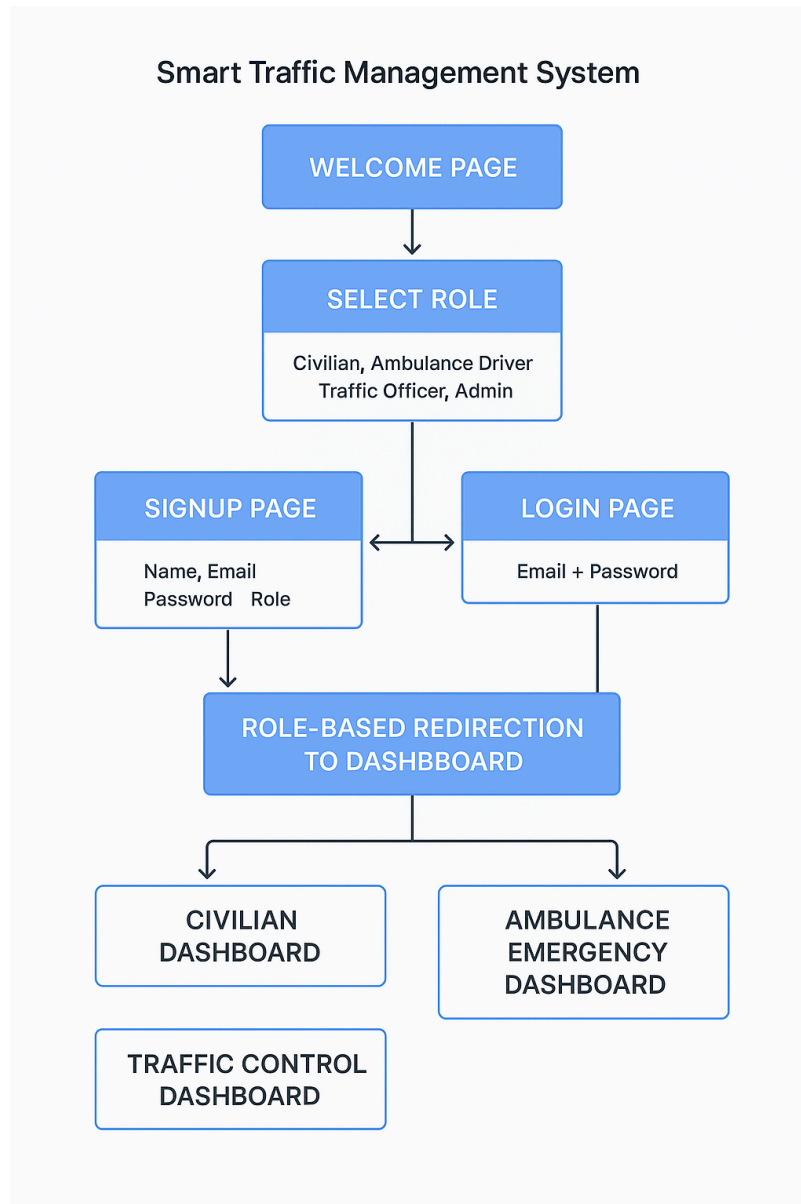
Description

- **Civilian Login:** Allows general users to access road-safety features such as pothole reporting, Tow help, and live transparency dashboards, enabling citizen participation in improving traffic conditions.
- **Ambulance Driver Login:** Provides verified emergency vehicle drivers with access to the Emergency Signal Override System, enabling real-time GPS tracking and automatic green-corridor support during emergencies.
- **Traffic Officer Login:** Gives authorized traffic personnel full access to the smart intersection dashboard, including live video feeds, lane-density monitoring, incident management, and signal-control tools.
- **Admin Portal Login:** Reserved for system administrators who manage users, intersections, device provisioning, data oversight, and overall platform configuration to maintain secure and smooth operations.

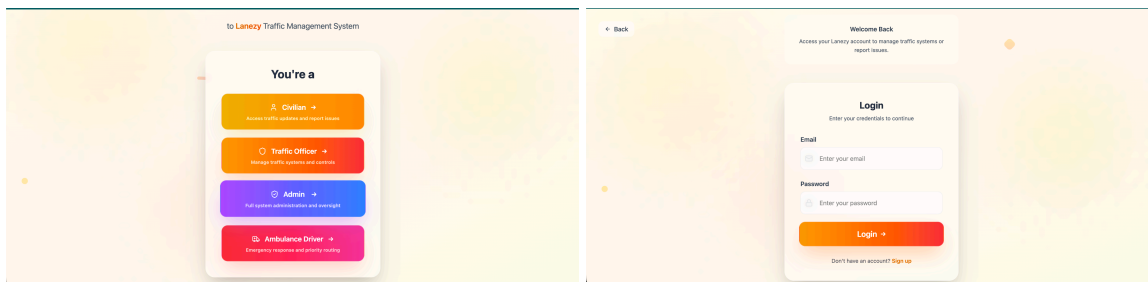
How to Use

- Select your role (Civilian, Admin, Ambulance Driver, or Traffic Officer) from the Welcome screen.
- Click **Sign Up** and fill in your name, email, password, and role.
- Submit the form to create your account and get registered in the system.
- After signup, proceed to the **Login** page to access your role-based dashboard.

FlowChart



Images



2. PRR-MASC(Smart Traffic Routing):

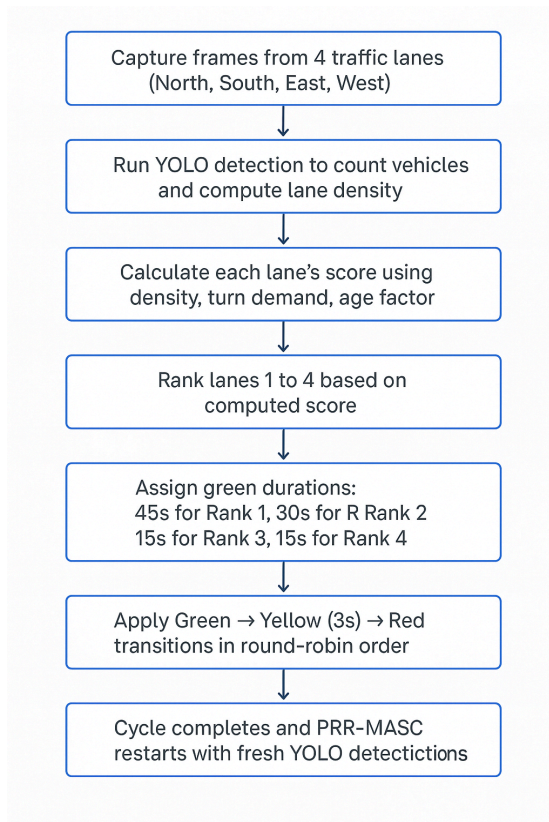
Description:

1. PRR-MASC (Priority Round-Robin Movement-Aware Signal Control) is a lightweight, engineered traffic algorithm that ranks all four lanes based on real-time YOLO density, turning behavior, and lane waiting age to ensure fairness and efficiency without starvation.
2. Unlike A2C reinforcement learning, PRR-MASC is deterministic, predictable, and ready for live deployment from day one, requiring no training dataset, simulation, or GPU resources.
3. The system captures YOLO detections, computes lane scores, ranks all lanes, and cycles through them in a fixed round-robin order, assigning green times of 45s, 30s, and 15s with a 3-second yellow transition.
4. Age factor boosts lanes that have waited longest, ensuring every lane gets serviced each cycle and eliminating starvation, while still prioritizing heavier traffic demand.
5. Here is the reference document supporting our decision model and theoretical comparison: <https://arxiv.org/pdf/2109.00937> (A2C vs. Rule-Based Adaptive Systems).

How It Works:

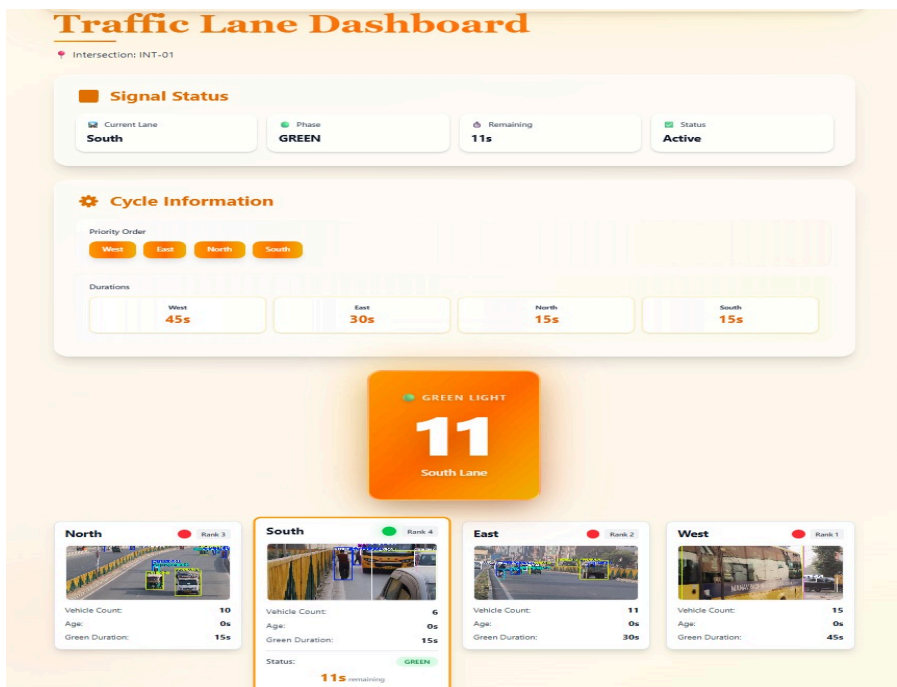
1. The system captures frames from all four cameras simultaneously and runs YOLO to count vehicles for each lane in real time.
2. Each lane is assigned an “effective load” score using density + turn probability + waiting age.
3. Lanes are ranked from highest to lowest score, and this rank defines the round-robin order for the cycle.
4. Traffic signals switch lane-by-lane in this fixed order using the durations 45s → 30s → 15s → 15s, with a 3s yellow between transitions.
5. After the full cycle completes, PRR-MASC re-evaluates the lanes using new YOLO detections and repeats the process for the next optimized cycle.

Flowchart



Images





3. Wrong Side Detection:

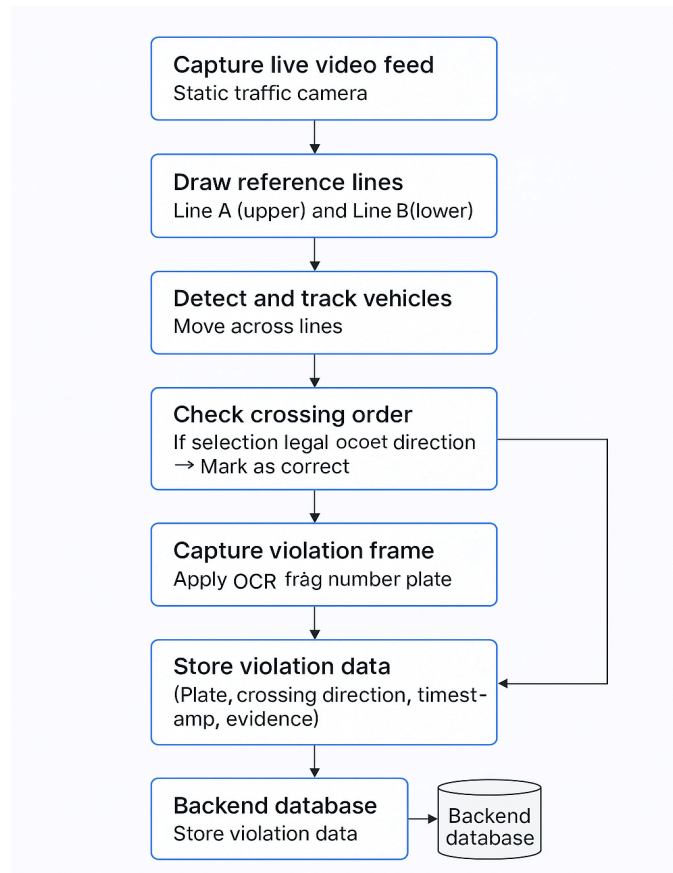
Description:

1. The wrong-side detection system identifies vehicles moving in the opposite direction by analyzing their movement between two configurable horizontal reference lines drawn on the video feed.
2. The system supports both possible legal directions — A→B or B→A and the traffic officer can select the correct direction for each road or camera setup.
3. Any vehicle crossing the lines in the reverse order of the selected legal direction is instantly flagged as a wrong-side violation.
4. OCR automatically extracts the number plate of the violating vehicle, and all violation details are stored in the backend for challan generation and traffic enforcement.
5. The solution is lightweight, deterministic, fast, and deployable in real-time without requiring heavy machine learning models.

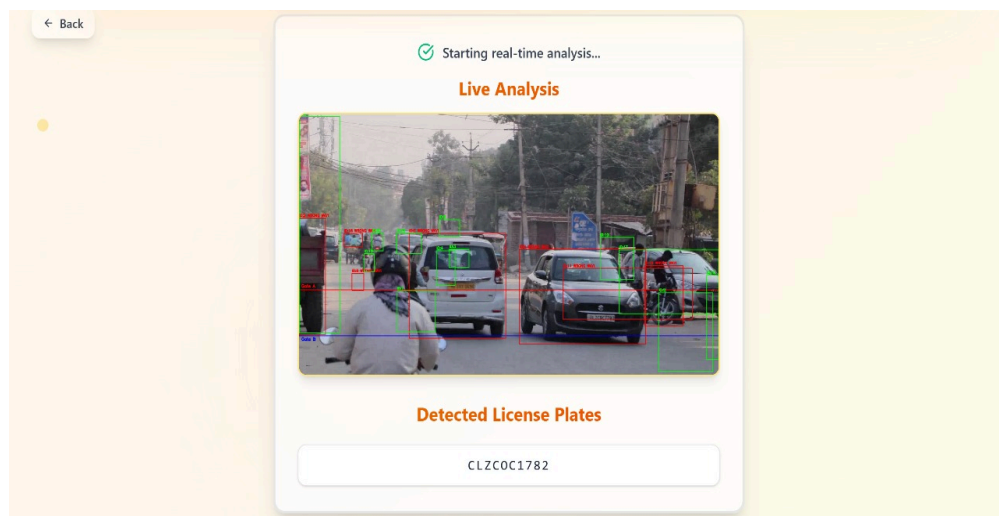
How it works

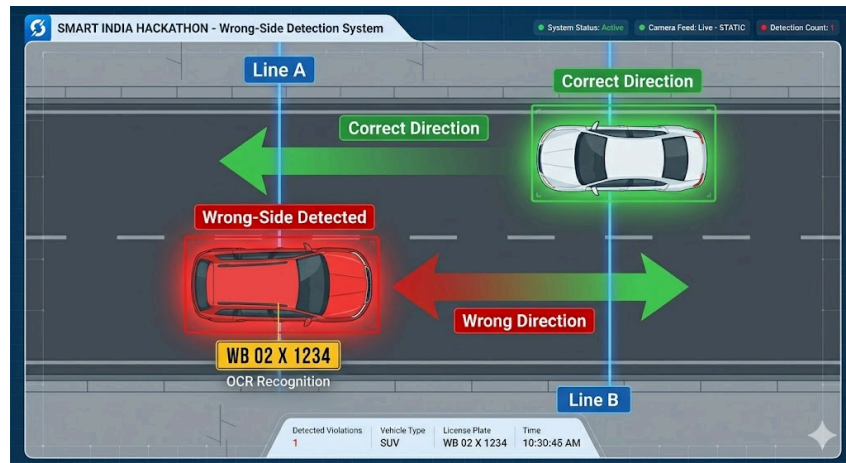
1. Two virtual horizontal lines (Line A and Line B) are placed on the live camera feed, defining the movement direction to be monitored.
2. The officer selects the legal movement direction — either A→B or B→A — based on the actual flow of that road.
3. Each detected vehicle is tracked to determine whether it crosses the lines in the chosen legal order or in the opposite, illegal order.
4. If the crossing order contradicts the selected legal direction, the system flags the vehicle as a wrong-side violator.
5. A cropped frame is captured, OCR extracts the number plate, and the complete violation record (plate, direction, timestamp, evidence) is saved to the database.

FlowChart



Images





4. Emergency Vehicles

Description

- The Emergency Vehicle Override System enables authenticated ambulance drivers to clear traffic automatically without manual signal control. When an ambulance driver presses the **Start Emergency** button on the mobile dashboard, the system begins tracking the ambulance's **GPS location**, **movement speed**, and **heading direction** in real time.
- Using these parameters, the system intelligently identifies the **nearest upcoming intersection** (not just any closest one) by analyzing both **distance and direction of travel**. Once the correct intersection is identified, the system determines the correct lane (North / South / East / West) that the ambulance will approach.
- The PRR-MASC cycle is temporarily paused, and the system forces the selected lane to **GREEN** until the ambulance crosses the intersection. After crossing, the override is automatically removed, and the traffic controller resumes normal PRR-MASC operation. This ensures both **rapid emergency clearance** and **balanced traffic fairness** for other lanes.
- The entire process is **automated**, **hardware-free**, **scalable**, and works using standard smartphone sensors only.

How It Works

1. Live GPS and Heading Capture

As soon as the Emergency Dashboard opens, the browser continuously collects latitude, longitude, and speed using the Geolocation API. Heading direction is calculated using two consecutive GPS points through the standard bearing formula used in Google Maps.

2. Stabilization Phase (5 Seconds)

The system collects data for 5 seconds to stabilize any fluctuations from stationary sensors or GPS jumps. After averaging multiple readings, the final heading direction is locked.

3. Emergency Trigger (Start Button)

When the driver presses Start Emergency, the system immediately uses the latest GPS + heading + speed data to analyze all intersections stored in the database.

4. Nearest Forward-Facing Intersection Selection

Using the Haversine distance formula and heading-angle comparison, the system identifies the next intersection in the direction of travel (avoiding incorrect selection of a past intersection).

5. Lane Direction Mapping

The selected intersection's reference lanes (north/south/east/west) are mapped to the ambulance's heading direction to determine exactly which lane should turn green.

6. Override of PRR-MASC Logic

The normal PRR cycle is paused, and the selected lane is forced to GREEN. All other lanes turn RED to create a green corridor for the ambulance.

7. ETA Tracking and Movement Monitoring

Dynamic ETA is calculated as:

$$\text{ETA} = \text{distance_to_intersection} / \text{ambulance_speed}$$

The system keeps updating the ambulance's position in real time.

8. Auto-End When Intersection is Crossed

When the ambulance moves beyond the intersection point (distance begins increasing again), the system automatically:

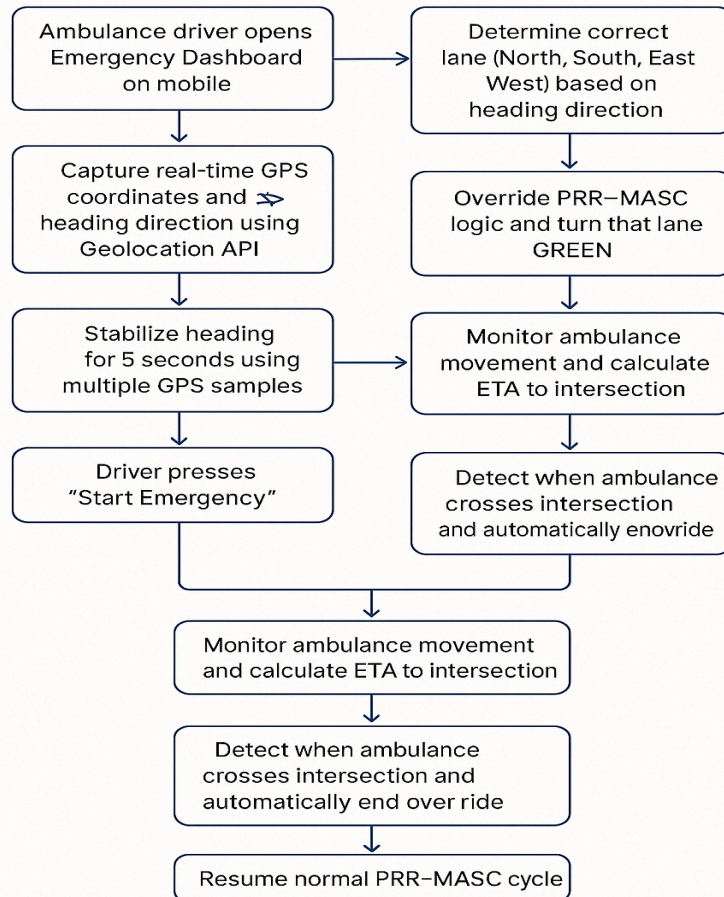
- Ends override
- Logs the event
- Restarts PRR-MASC from a fresh cycle

9. Full Logging for Audit and Safety

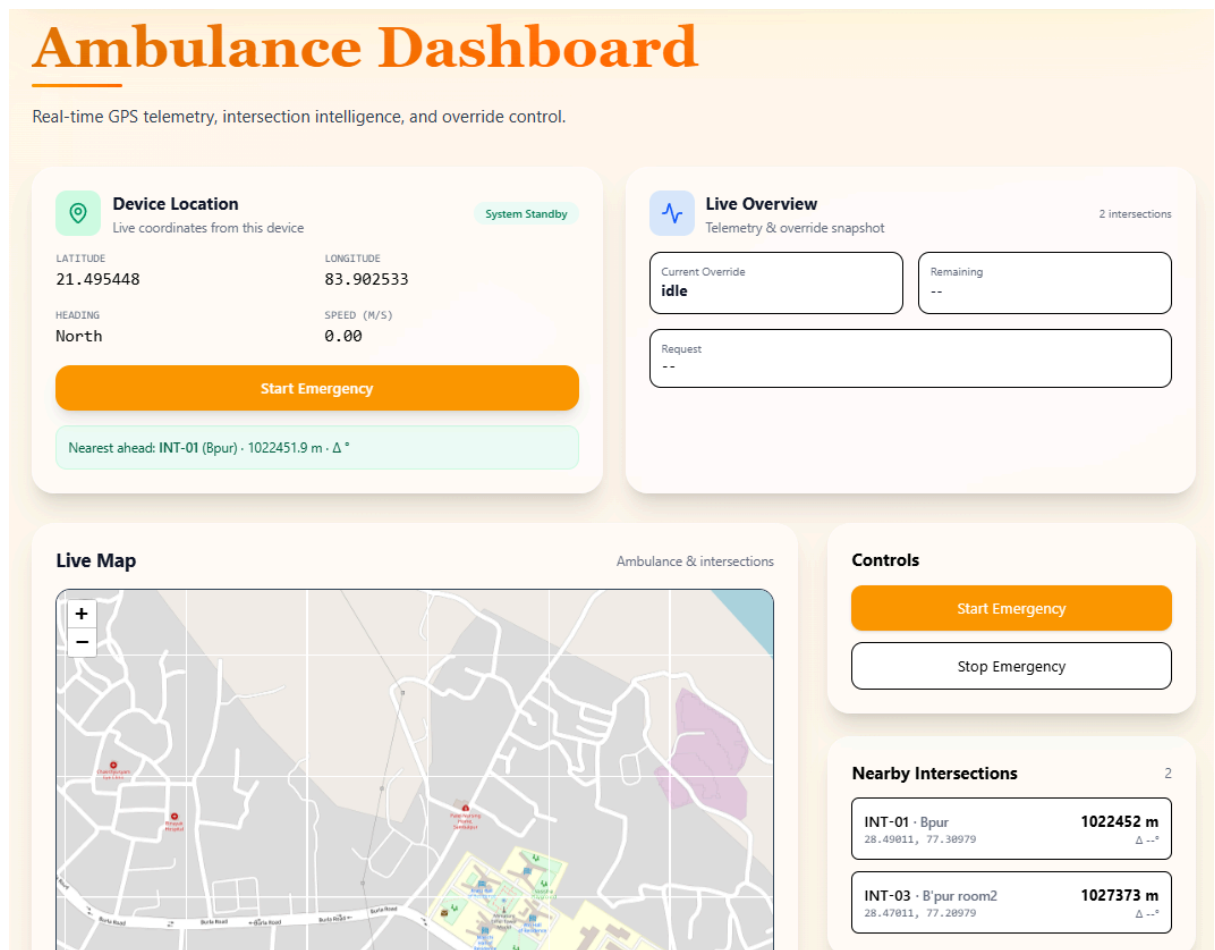
Driver ID, location, intersection selected, lane forced, time taken, and override status are stored for officer review.

FlowChart

EMERGENCY VEHICLE AUTOMATIC SIGNAL OVERRIDE SYSTEM



Images



5. Pothole Detection and Reporting

Description

- The Laneasy pothole detection module allows both civilians and government IoT devices to detect and report potholes accurately. Civilian users can take a real-time camera photo of a pothole and upload it. The system verifies the authenticity of the image using EXIF metadata, GPS location, timestamp, and live camera-only restrictions to prevent fake reporting or “photo-of-photo” misuse.
- In parallel, a dedicated IoT device inspired by Ather's pothole sensing system (using accelerometer + gyroscope + GPS) is deployed on government vehicles to automatically detect strong road depressions and transmit coordinates to the backend. This builds a larger, richer dataset for pothole prioritization.
- All pothole detections—civilian and IoT-based—are plotted on a dynamic digital pothole heatmap, visible to both civilians and

authorities, ensuring complete transparency about which potholes are detected, pending, or resolved.

How it works

1. User selects "Report Pothole" and the camera opens; gallery upload is disabled to prevent uploading old or fake images.
2. When the picture is captured, EXIF metadata is extracted: timestamp, GPS, camera make/model, orientation, and capture mode.
3. YOLO detection runs on the live-captured frame to check if a pothole exists.
4. The server verifies that EXIF GPS $\approx$ browser's geolocation to ensure the photo was taken on the spot.
5. If verified, the system maps the location to a 50m grid cell and increments pothole count for that segment.

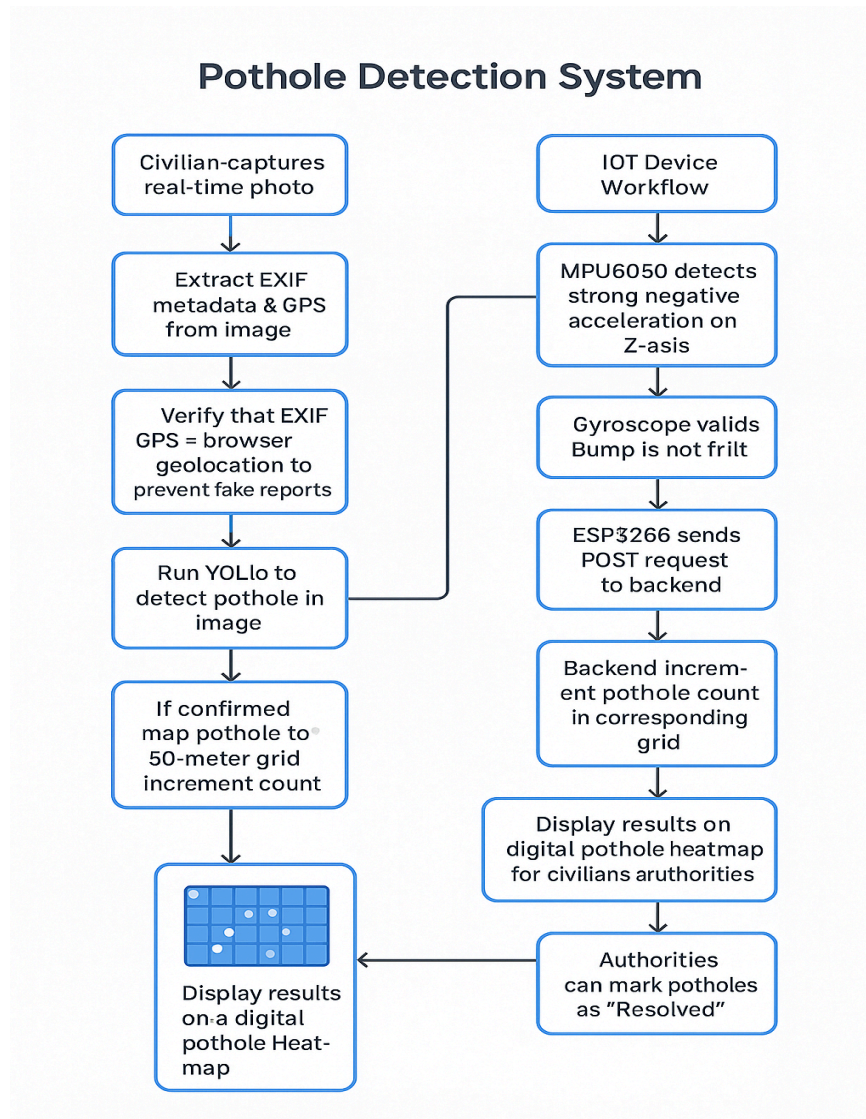
IoT Ather-Style Device Detection

1. An IoT module with MPU6050 (accelerometer + gyroscope) + NEO-6M GPS + ESP8266 is mounted on government vehicles.
2. The accelerometer detects sudden negative Z-axis drops (pothole signatures) while filtering out slow bumps like speed breakers.
3. Gyroscope readings verify if the dip is real or caused by phone/device tilt.
4. GPS coordinates are attached and sent as a POST request to the Laneasy backend.
5. The backend increments the pothole count for the nearest 50m road segment.

Pothole Transparency Map

1. Every pothole detection is logged with count, location, confidence score, reporter type (civilian/IoT), timestamp, and status.
2. The map visualizes pothole severity via color-coded heatmaps (green $\rightarrow$ moderate $\rightarrow$ red).
3. Authorities mark potholes as "Resolved," and civilians can see transparency of action taken.

FlowChart



Images



6. Illegal Parking System

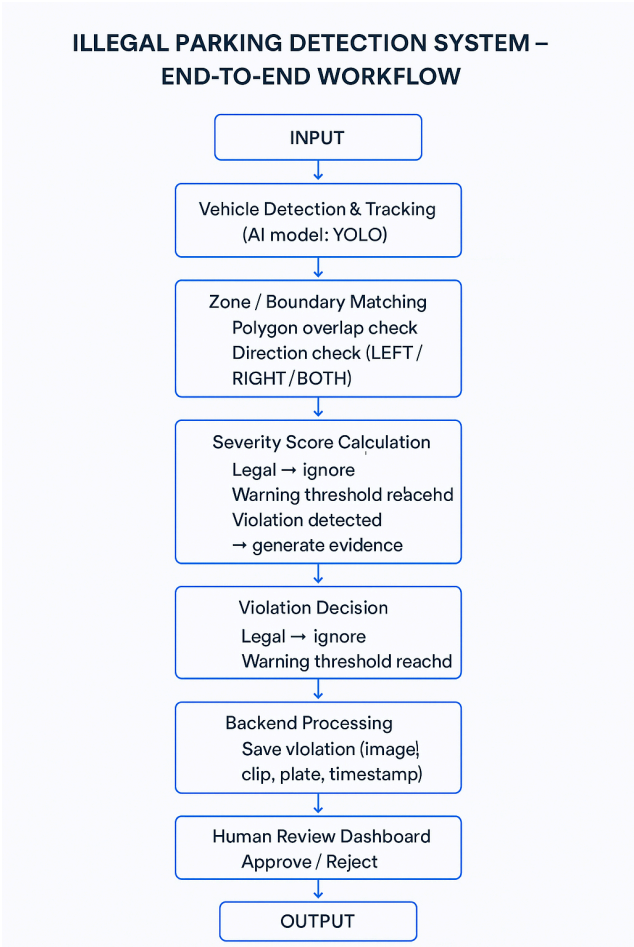
Description

Illegal Parking Detection System automatically identifies vehicles that stop or park in prohibited zones using CCTV feeds, AI-based object tracking, and geo-configured rule zones. Administrators define no-parking polygons or directional boundary lines through an interactive UI. The system continuously monitors dwell time, lane blockage, and direction of violation using on-edge AI processing. Violations are logged with vehicle image, number plate, timestamp, severity, and lane impact, and sent for human verification before challan generation. This drastically improves lane availability, reduces congestion, and enables real-time enforcement at scale.

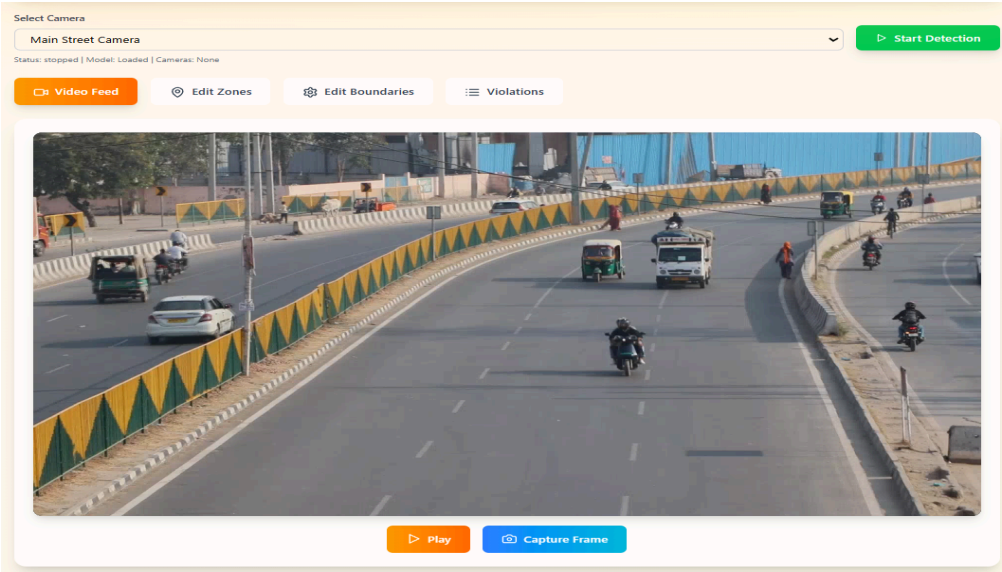
How it works ?

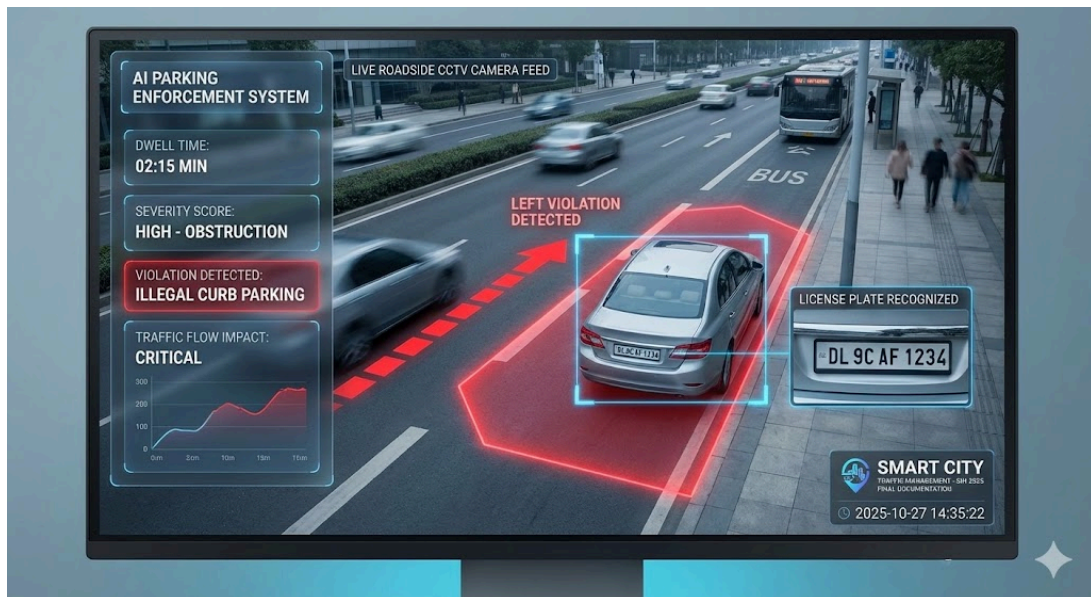
- **Camera Feed → AI Model:**
Live footage from roadside cameras is processed using YOLO-based detection to identify vehicles and track movements.
- **Custom Zones / Boundary Lines:**
Administrators draw polygons or directional lines in the UI to define no-parking, bus lanes, footpaths, or illegal sides of the road.
- **Violation Logic:**
The engine computes whether a vehicle is inside the polygon or on the illegal LEFT/RIGHT side of a boundary line and remains stationary beyond a threshold.
- **Severity & Evidence:**
System calculates blocked-lane ratio, dwell time, and severity score; captures vehicle image, plate, timestamp, and optional video clip.
- **Review & Enforcement:**
Violations go to a dashboard where traffic officials approve or reject them; approved cases auto-trigger challan generation or towing request.

FlowChart



Images





7. Red Light Jumping

Description:

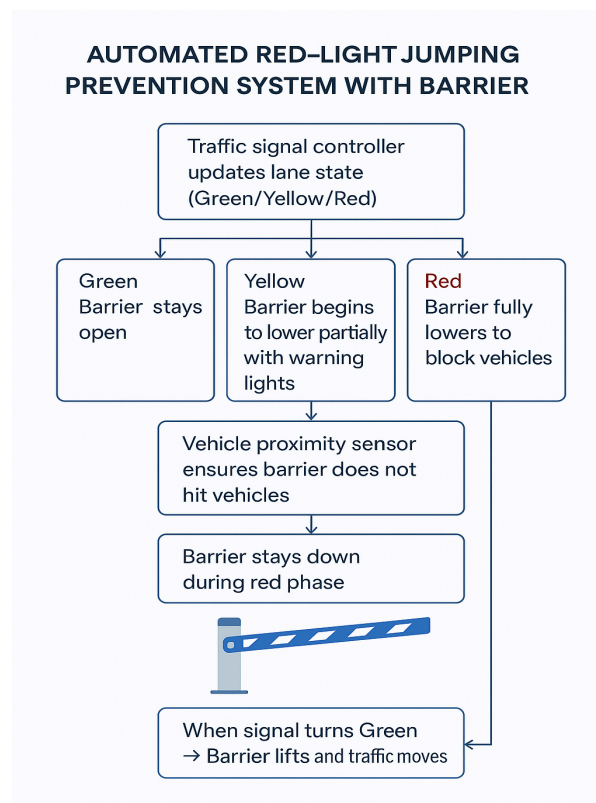
- Traditional red-light enforcement relies purely on cameras and backend challan systems, which often fail due to delayed notifications, plate detection errors, low light, spoofing, and absence of real-time driver deterrence. Laneasy introduces a physical barrier enforcement system, similar to toll plaza bollards, where a lightweight automatic banner/gate drops down whenever the signal turns red.
- When the signal is GREEN, the barrier stays open.
When the signal turns YELLOW → RED, the barrier smoothly comes down and physically prevents a vehicle from jumping the signal.
- This system does not rely on challans, does not require OCR, and works instantly through physical enforcement, reducing violations dramatically.

How it works?

1. The traffic signal controller (PRR-MASC or manual signal) sends real-time state updates (Green/Yellow/Red) to the barrier actuator system installed near the stop-line.

2. When the lane turns GREEN, the barrier remains open and vehicles pass normally.
3. As soon as the lane transitions to YELLOW, the system partially lowers the barrier and flashes a warning signal to alert drivers that red is coming.
4. When the lane turns fully RED, the actuator lowers the banner/gate completely, preventing vehicles from crossing the stop line physically.
5. Sensors detect if a vehicle is too close when lowering; the system pauses to avoid hitting vehicles and resumes closing when safe.
6. The barrier remains closed for the entire red duration, enforcing perfect compliance without requiring challan cameras.
7. Once the green phase resumes, the system lifts the barrier smoothly and resets for the next cycle.
8. The actuator logs every lowering/lifting event to the backend, giving authorities complete transparency.

Flowchart



Images



8. Tow Services

Description

Breakdown incidents on Indian roads often leave drivers helpless because they do not know their exact location, local tow services, who is available, or which service is trustworthy. Laneasy solves this by introducing an **instant, location-aware breakdown assistance system** for buses, autos, private vehicles, taxis, and two-wheelers.

When the driver clicks the **Breakdown SOS** button, the system automatically detects their **precise GPS location**, identifies the **nearest 10 tow/troubleshooting services**, and sends an instant request to all of them simultaneously through mobile/internet notifications.

The **first service provider to accept the request** is assigned the job. The driver receives an SMS/notification with:

- The tow service name
- Contact details
- ETA
- Live-moving map

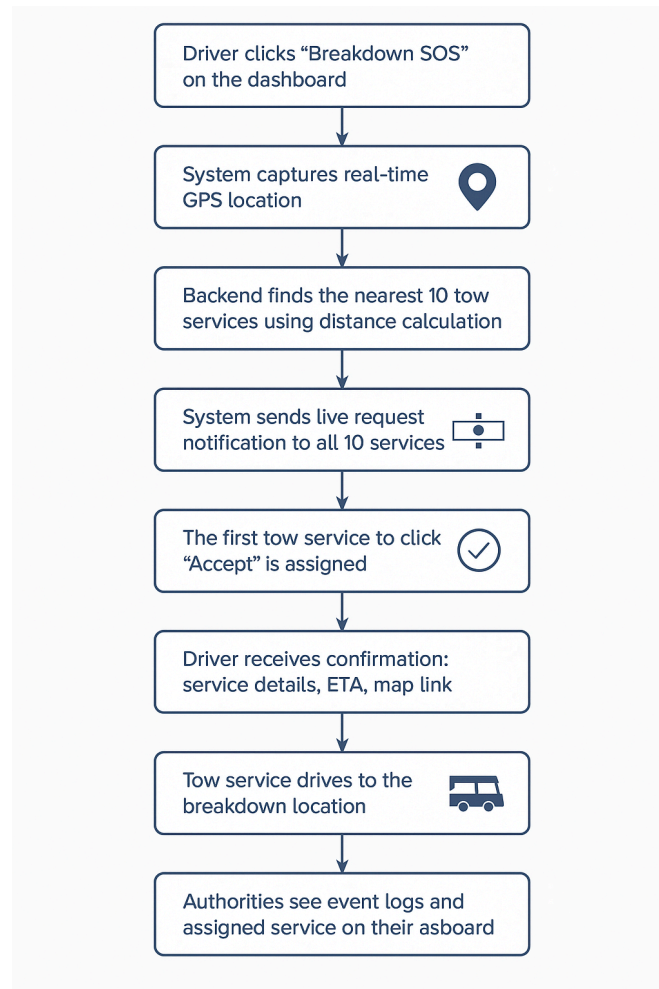
- Verification badge

This makes breakdown handling **fast, transparent, reliable, and panic-free**.

How it Works ?

1. The driver logs into the Civilian Dashboard and clicks the “Breakdown SOS” button during vehicle failure.
2. The system immediately captures the driver's live GPS using the Geolocation API, even if they don't know their location.
3. The backend calculates and sorts all registered tow services by distance (Haversine formula).
4. The nearest 10 tow services receive an instant Breakdown Request Notification with:
 - driver location
 - distress type
 - timestamp
 - route map
5. The first tow provider to press “Accept Request” gets assigned to the breakdown event. Others are automatically blocked from accepting.
6. Driver receives an SMS + on-screen confirmation with the assigned tow provider's:
 - Name
 - Phone
 - Live location
 - ETA
 - Vehicle number
7. Authorities can see breakdown events live on their dashboard for verification, support, or emergency coordination.
8. After job completion, the system logs the event for transparency and reliability tracking of tow services.

FlowChart



Images

